

Electrodynamics

Independence Semester 2024

Homework 2

September 14, 2024

Problem 1

a) An electron (e^-) with energy (E_e) of 20 GeV is colliding head-on with a proton (p) with energy (E_p) of 245 GeV. You can consider the electron and proton to be effectively mass-less since $E_{e/p} \gg m_{e/p}$. First, find out an expression for the energy in the center-of-momentum (CM) frame of the e^-p system involving E_e and E_p . Next, evaluate the numerical value of the CM energy of the e^-p system.

[Caution: You must use the 4-vector notation for this problem.]

b) Let us consider a short-lived particle of mass m moving with a relativistic velocity v in the positive x -direction. It decays in a time τ in its own rest frame (CM frame). Where does the decay take place in the Laboratory frame and when does it take place as measured by our laboratory clocks?

Problem 2

a) Show that if a (real) Cartesian vector V^i in 3-dimensional Euclidean space satisfies $V^i V^i = 0$, then it must be that $V^i = 0$.

b) Show that if a (real) 4-vector V^μ in 4-dimensional Minkowski space-time satisfies $V^\mu V_\mu = 0$, then V^μ need not be zero.

c) A particle following the worldline $x^\mu = x^\mu(\tau)$ has 4-velocity $U^\mu = \frac{dx^\mu}{d\tau}$ and 4-acceleration $a^\mu = \frac{dU^\mu}{d\tau}$, where τ is the proper time. Show that $U^\mu a_\mu = 0$.

Problem 3

- a) Suppose we know that $\phi \equiv A^\mu B_\mu$ is a Lorentz scalar, where A^μ is an arbitrary 4-vector. Prove that B_μ must therefore be a 4-vector. (This is a special case of a more general quotient theorem for tensors. Note that it is essential in the argument that A^μ can be an arbitrary 4-vector; make sure your proof makes use of this!)
- b) Show that the tensor $S_{\mu\nu} = W_{\mu\rho} W_\nu^\rho$ is always symmetric, regardless of any symmetry properties of the tensor $W_{\mu\nu}$.

Problem 4

- a) Define the scalar quantity R by $R^2 = \eta_{\mu\nu} x^\mu x^\nu$. Show that $\partial_\mu R = \eta_{\mu\nu} x^\nu / R$.
- b) Hence show that $\square \frac{1}{R^2} = 0$ (assuming $R \neq 0$).
- c) Derive the condition on the constant 4-vector k_μ such that $\phi \equiv e^{ik_\mu x^\mu}$ solves the wave equation $\square \phi = 0$.

Problem 5

- a) An electrostatic potential near the origin of an observer $\mathcal{O}$ is given by $\phi(x, y, z) = \frac{V_0}{a^2}(y^2 + z^2)$. The absence of magnetic field implies the corresponding vector potential $\vec{A} = 0$. Find the charge density ρ near the origin (in terms of input parameters V_0, a).
- b) An observer $\mathcal{O}'$ slides with velocity v along the overlapping x, x' axes. The other axes are parallel, $y||y', z||z'$. As $(\phi, \vec{A})$ transforms into $(\phi', \vec{A}')$ like $(t, \vec{x})$ into $(t', \vec{x}')$, determine ϕ' and $\vec{A}'$.
- c) Find the electric and magnetic fields observed by $\mathcal{O}'$.
- d) Find the corresponding charge density ρ' , and the current density $\vec{J}'$.

Problem 6

The transformations of $\vec{E}$ and $\vec{B}$ under a general Lorentz boost with velocity $\vec{v}$ are

$$\begin{aligned}\vec{E}' &= \gamma(\vec{E} + \vec{v} \times \vec{B}) - \frac{\gamma - 1}{v^2}(\vec{v} \cdot \vec{E})\vec{v} \\ \vec{B}' &= \gamma(\vec{B} - \vec{v} \times \vec{E}) - \frac{\gamma - 1}{v^2}(\vec{v} \cdot \vec{B})\vec{v}\end{aligned}\tag{1}$$

Using the above relations show explicitly that the following quantities remain invariant under Lorentz transformation

a) $\vec{B}^2 - \vec{E}^2$,

b) $\vec{E} \cdot \vec{B}$.

[Caution: You must use the summation notation for your calculations of this problem.]

Problem 7

a) Suppose that constant, uniform $\vec{E}$ and $\vec{B}$ fields are orthogonal in the frame S . Look for a Lorentz boost using a velocity orthogonal to $\vec{E}$ and $\vec{B}$, i.e. of the form $\vec{v} = \lambda \vec{E} \times \vec{B}$, where λ is a constant that you should determine, such that in the frame S' the magnetic field vanishes, $\vec{B}' = 0$.

b) Bearing in mind that any boost velocity must be less than the speed of light, find the condition involving $\vec{E}$ and $\vec{B}$ that must be satisfied in order for the boost in part a) to be possible.

c) Under what circumstances (expressed again as a condition involving $\vec{E}$ and $\vec{B}$) is it impossible to find a boost of the form discussed in this question that allows one or other of $\vec{E}'$ and $\vec{B}'$ to be set to zero?

Problem 8

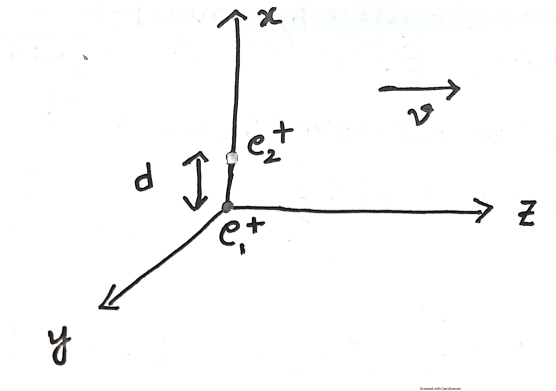


Figure 1: Figure for problem 4

Consider two highly relativistic positrons ($\gamma \sim 10^5$) in an positron beam. Initially, in the beam frame they are separated by a distance d , and the beam is moving in the

positive z -direction with an uniform velocity v . The positrons e_2^+ and e_1^+ are travelling side by side as shown above.

- a) Write down expressions at e_1^+ giving the effect of e_2^+ in the rest frame. Specifically evaluate the following vectors: $\vec{E}$, $\vec{B}$, the Lorentz force $\vec{F}$, and the acceleration $\vec{a}$.
- b) Evaluate the same vectors in the laboratory frame. The results will differ by various relativistic factors (e.g. γ).
- c) Give intuitive explanation of these factors.